

APPROVED METHOD:
PHOTOMETRIC TESTING OF INDOOR LUMINAIRES
USING HIGH INTENSITY DISCHARGE OR
INCANDESCENT FILAMENT LAMPS
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-46-20(R2023)

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Publication of this document
has been approved by the IES.
Suggestions for revisions
should be directed to the IES.

**Prepared by:
The IES Testing Procedures Committee**



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Approved by the IES Standards Committee, December 19, 2022, as a Transaction of the Illuminating Engineering Society.

Approved April 28, 2023, as an American National Standard.

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Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

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Printed in the United States of America.

ISBN# 978-0-87995-211-2

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1.0 Introduction and Scope

1.1 Introduction

This Approved Method is intended to promote uniform test results as obtained under controlled laboratory conditions in measuring the luminous flux and intensity of high intensity discharge (HID) and incandescent indoor-type luminaires.

This document is a revision of and replacement for LM-46-04 (Reaffirmed 2012), *IESNA Approved Method for Photometric Testing of Indoor Luminaires Using High Intensity Discharge or Incandescent Filament Lamps*.

1.2 Scope

The photometry of luminaires with high intensity discharge (HID) lamps, including mercury vapor, high-pressure sodium (HPS), or metal halide lamps, is covered by this method. Also covered are similar luminaire types employing incandescent lamps, including tungsten-halogen lamps. For the unique testing requirements of entertainment lighting luminaires (i.e., for stage, TV, or film) using HID and incandescent lamps, the reader should consult *ANSI/IES LM-73-18, Approved Method: Photometric Testing of Entertainment Lighting Luminaires Using Incandescent Filament or High-Intensity Discharge Lamps*.¹ Because of their special characteristics, low-pressure sodium lamps and fluorescent lamps are not discussed in these procedures. For information on fluorescent lamp testing, the reader should consult *ANSI/IES LM-41-20, Approved Method: Photometric Testing of Indoor Fluorescent Luminaires*.²

2.0 Selection of Lamps

The lamps selected shall conform to the lamp manufacturer's nominal design dimensions and parameters. Lamps with phosphor coatings shall be carefully selected to obtain a phosphor density uniform over the entire lamp envelope. It is important to ensure that the centering of the arc tube and other mechanical details are within the limits recommended by the American National Standards Institute (ANSI).³

3.0 Selection of Luminaires

Each luminaire selected should be representative of the manufacturer's typical product. The luminaire shall be clean, except in cases where dirt depreciation measurements are the reason for the photometry. Precise location of the lamp within the luminaire can be extremely important, particularly in the case of beam forming equipment.

4.0 Test Conditions

4.1 Extraneous Light

Extraneous light is any light that reaches the photodetector other than directly from the source being measured (or reflected from the mirror of a moving-mirror photometer) because of extraneous reflections or other light sources. Adequate precautions shall be taken to minimize such reflections by using flat black surfaces and by using angles, grooves, baffles, and other light traps.

Extraneous light can be measured by running a complete or partial test with the direct light from the luminaire completely shielded from the photodetector. The photodetector then sees only extraneous light, which can be measured, recorded, and then subtracted during the calculation of the luminaire luminous intensities. Care shall be taken to block only direct light while simultaneously providing minimal interference to extraneous light paths.

4.2 Ambient Temperature

For precise measurement of photometric and electrical characteristics of luminaires with HID and incandescent lamps, the ambient temperature should be maintained at $25\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F} \pm 9\text{ }^{\circ}\text{F}$). This temperature shall be measured at a point not more than 1.5 meters (5 feet) from the lamp or luminaire and at the same height as the lamp or luminaire. The temperature-sensing device shall be shielded from direct radiation of the light source. Any deviations from the recommended temperature specifications shall be noted in the report.

4.3 Air Movement

The luminaire (or test lamp during calibration) shall be tested in relatively still air. A maximum airflow of 0.08 meters per second (16 feet per minute) is suggested.

4.4 Lamp Seasoning

Test lamps shall be seasoned such that their characteristics remain constant during the test to be conducted. (Refer to *IES LM-54-12, IES Guide to Lamp Seasoning* for detailed information.⁴⁾

4.5 Test Lamp Operation

Accurate measurements of light require that the lamp be operated on the proper electrical circuits. Stable performance is of primary importance. The lamp is considered stabilized when the monitoring of light output over a period of 30 minutes produces fluctuations no greater than $\pm 1.0\%$ between any two consecutive readings taken approximately 15 minutes apart. Care shall be taken to ensure that successive light readings are not increasing or decreasing monotonically. A false sense of stabilization can occur if the sampling period is short compared to the normal lamp stabilization time. A stabilization time of ten minutes is usually sufficient for incandescent and tungsten-halogen lamps, but may range from one to three hours for mercury vapor, HPS, and metal halide lamps.

HID lamps operating on magnetic ballasts shall be operated within $\pm 2.0\%$ of the rated power of the lamp. HID lamps operating on electronic ballasts shall have the ballast input voltage set to within $\pm 0.5\%$ of the unit's rated primary voltage; if the voltage is a range, the nominal voltage should be used and indicated on the test report.

Tungsten-halogen lamps shall be operated at a constant value of current near their nominal voltage.

Other incandescent filament lamps may be operated at a constant current that corresponds to 90% of rated input lamp voltage (this practice extends the life of test lamps). When conducting absolute photometry, lamps should be operated at rated voltage. Operation at other voltages should be specifically noted in the test report.

For a test in the field, it is recognized that circuit voltage (rather than lamp power) will probably be adjusted to nominal. In this case, special attention shall be paid to thermal and electrical effects that different ballasts can have on a lamp and its light output. (Refer to *ANSI/IES LM-61-20, Approved Method: Identifying Operating Factors for Installed Outdoor High Intensity Discharge (HID) Luminaires*⁵ and other industry guides for specific information about special factors and how to produce and apply them.) A special testing procedure may be required to correlate field results with laboratory results.

For more detailed information on electrical measurements the reader should consult *ANSI/IES LM-51-20, Approved Method: Electrical and Photometric Measurements of High Intensity Discharge Lamps*⁶ and *ANSI/IES LM-45-20, Approved Method: Electrical and Photometric Measurements of General Service Incandescent Filament Lamps*.⁷

4.6 Test Distance

The distance between the test lamp or luminaire and the photodetector shall be at least five times the maximum dimension of the luminous opening of the luminaire, and a minimum of 3 meters (10 feet). The test distance shall be recorded in the test report.

4.7 Electrical Requirements

4.7.1 Power Source Voltage Regulation. The root-mean-square (RMS) voltage or current of the AC power source shall be regulated to within $\pm 0.5\%$.

4.7.2 Power Source Wave Shape. The AC power source shall have a sinusoidal voltage wave shape such that the RMS summation of the harmonic components does not exceed 3.0% of the fundamental.

4.8 Instrumentation

Electrical instruments shall be selected and used with care to ensure accurate measurements. For analog meters it is recommended that the test range be selected so that data for any specific measurement are taken between the midpoint and the maximum scale reading. Since gaseous-discharge lamps might have seriously distorted wave shapes, AC voltmeters and ammeters should be selected that respond to true RMS

values. Wattmeters should respond to average power. Instruments whose scales are calibrated in RMS values but whose readings depend on peak or average values shall not be used. Instrument indications should have good reproducibility. Lamp current or power, depending upon the type of light source, shall be checked with a calibrated instrument that has an accuracy of $\pm 0.25\%$. The effect, if any, of the instruments on measured quantities shall be addressed (refer to *ANSI/IES LM-28-20, Approved Method: IES Guide for the Selection, Care, and Use of Electrical Instruments in the Photometric Laboratory* for detailed information⁸).

4.9 Goniophotometer

4.9.1 General. The goniophotometer provides a means of mounting the test sample and rotating it through the required angular traverses. The goniophotometer shall be rigid enough to provide correct measurement of angles even when there is an appreciable off-balance loading. The goniophotometer should provide for accurate angular settings that are reproducible to within $\pm 0.25^\circ$.

4.9.2 Goniophotometer Systems. For the photometric testing of indoor HID luminaires, a Type C goniophotometer shall be used. Type C goniophotometers are ideal for testing position-sensitive HID lamps. For indoor incandescent luminaires, if lamp orientation does not affect light output, Type A, B, or C goniophotometers may be used. However, regardless of the goniophotometer type, all data shall be reported in the Type C format (any Type A or Type B coordinate data shall be converted to its Type C coordinate equivalent).

For more details about goniophotometers and coordinate systems the reader is referred to *ANSI/IES LM-75-19, Approved Method: IES Guide to Goniophotometer Types and Photometric Coordinate Systems*.⁹ Annex A in that document explains how to determine the Type C coordinate equivalent of any Type A coordinate or any Type B coordinate.

4.10 Photodetectors

An optically-corrected silicon cell shall be used as a photodetector because it exhibits low noise,

appropriate sensitivity, excellent stability and linearity, fast response, and low temperature sensitivity. The use of photodetectors and suitable measuring circuits are described elsewhere.^{8,10,11} The photodetector spectral response shall be such that $f_1' < 2.0\%$. (Refer to CIE S 023/E for more information concerning the factor f_1' .¹²)

5.0 Preparation of Equipment for Test

5.1 Luminaire Positioning on a Goniophotometer – General

The luminaire shall be mounted in a way that prevents supports or other adjacent objects from obstructing the photodetector's access to direct light or reflecting any other light into the photodetector. Where a moving-mirror photometer is used, the photodetector shall be shielded to "see" only the image of the luminaire's optical assembly. This will help to minimize the extraneous light reaching the photodetector.

In the case of beam-forming equipment where the light beam might be aimed at various vertical angles, the photometric test shall be performed with the beam aimed at an angle of typical use (normally at nadir or horizontal). Test orientation shall be noted in the test report. It is important that HID lamps operate with their arc tubes in the same position during the bare lamp calibration and the luminaire photometric test. To a lesser extent, lamp orientation can also affect the light output of incandescent lamps, depending on details of filament design and processing. It is recommended that incandescent lamps be in the same position for both luminaire and bare-lamp measurements when possible.

5.1.1 Lamp Calibration. The light center of the test lamp (or the geometric center of the light centers of the test lamps) shall be mounted so that it is at the center of the goniophotometer.

5.1.2 Luminaire Mounting. The luminaire shall be mounted on the photometer so that the following referenced planes or parts of the luminaire coincide with the center of the goniophotometer:

Case 1, recessed and coffered luminaires: The opening in major ceiling plane.

Case 2, suspended or surface mounted luminaires (see **Figure 5-1**):

A: Direct, semi-direct, semi-indirect, and indirect classifications: The plane of major light emitting opening, if the light center of the lamp is placed within the enclosing reflector.

B: Direct-indirect and general diffuse classifications: To midpoint of upward and downward major light openings; if such openings do not exist, to the geometric center in a vertical direction.

Case 3, open luminaires with reflector lamp exposed: The face of the reflector lamp, except where the lamp interacts substantially with the reflector so as to significantly modify the lamp output. In these instances of interaction, the aperture opening is to be centered.

In all three cases, the lamp(s) and the optical structure shall be adjusted to ensure the greatest degree of output symmetry.

5.1.3 Lamp Orientation. When used in photometric tests, high intensity discharge (HID) lamps require special precautions. Long stabilization periods (up to several days) shall be allowed when the operating

orientation is changed. Even designated “universal operating” HID lamps require prolonged stabilization in their actual operating position before they can be used as test lamps. In practice, separate lamps should be maintained for each lamp orientation (base up, base down, or horizontal) and should be operated only in their assigned orientation. Lamp seasoning should also be done in the same orientation as the intended orientation during testing.

Furthermore, the orientation of a horizontally operating HID lamp about its horizontal axis is critical. To maintain proper orientation, the arc tube fill tip is always positioned at the 12, 3, or 9 o'clock position. In conjunction with the arc tube fill tip positioning, horizontal HID lamp orientation should be with the arc support rod directed toward zenith. If the lamp has dual support rods, it should be directed toward zenith and nadir.

As mentioned previously, it is important that HID lamps be operated with the arc tube in the same position during the bare lamp calibration and the luminaire photometric test. The manufacturer’s guidelines for lamp positioning for best stability and proper operation should always be consulted. It is suggested that the lamp base be marked so that the proper position can be verified any time the lamp is energized. Finally, metal halide HID lamps cannot be moved after turn-off until

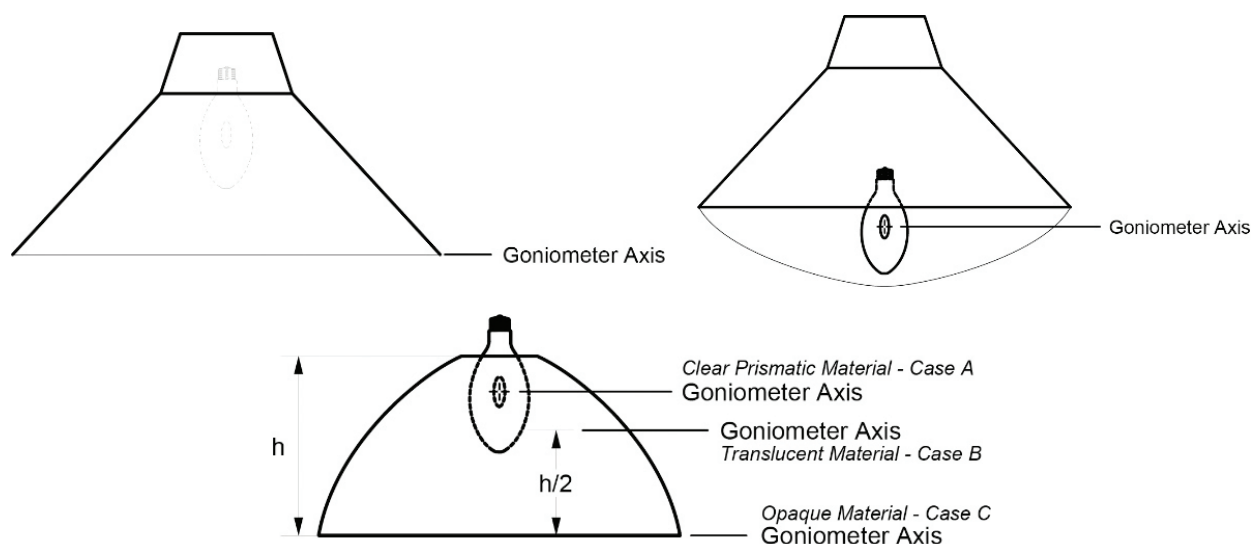


Figure 5-1. Positioning of a luminaire (suspended or surface-mounted) on the goniophotometer for three types of luminaires.

all salts within the arc tube have completely cooled and solidified. For most applications, 30 minutes of cooling is sufficient.

Single- and double-ended HID lamps without integral outer glass envelopes may require additional precautions and considerations. For example, certain such lamp types will not reach rated electrical, thermal, or photometric performance if operated in free air. A *luminaire simulator* will usually be required (see **Annex D** for typical construction details). When a luminaire simulator is used, lamp output may be reduced by as much as 3%. The actual reduction shall be determined and then corrected for in the final test report. Some HID lamps have quartz arc tubes and quartz outer envelopes that do not block ultraviolet (UV) radiation. Because luminaire simulators may not block UV radiation, precautions shall be taken to protect lab personnel from eye injury and actinic erythema (skin burning). Some HID lamps support ozone formation. Ozone is toxic and can cause health problems. The lamp manufacturer's guidelines should be consulted for specific information.

6.0 Test Procedure

6.1 Photometric Calibration

Photometric calibration relates the light output of test lamps to an assigned lumen value. If the luminaire is to be supplied with a specified manufacturer's lamp, initial rated lumens given by that manufacturer may be used. The test lamp should be calibrated in the same orientation it will assume when installed in the luminaire under test. It is necessary to correct for light output changes that occur if the lamp position within the luminaire does not agree with the lamp position used during photometer calibration. This is quantified by determining a *correction factor* whenever there is a position change. This correction factor might be obtained by placing the lamp in an integrating sphere, measuring its light output in the two applicable orientations, and then taking the ratio thereof.

6.1.1 Relative Method Photometry. The relative method of photometry is recommended for the great majority of HID and incandescent-filament luminaire

luminous intensity distribution tests. This is because the additional steps of using a calibrated, absolute lamp (or lamps) can be avoided.

First, the luminous intensity distribution of the bare lamp (that the luminaire will be equipped with) is read alone and recorded. For vertical lamps, this is achieved by measuring the lamp luminous intensity distribution in a series of 10 or more vertical half-planes (vertical angles 0 to 180 degrees) spaced equally around the luminaire circumference. Tests also need to be performed for horizontal lamps. A minimum of 16 half-planes should be taken to accurately describe the luminous intensity distribution for horizontal lamps. When taking bare-lamp readings, measurement planes that include shadows from arc tube support wires, lamp socket bases, or the goniophotometer support structure shall be avoided. Typically, horizontal lamps are offset half of the lateral test increment (the horizontal angle between measured planes) when tested to avoid measuring planes that may be blocked by the goniophotometer support structure or lamp socket.

Within each vertical half-plane, intensity is measured at 36 vertical angles with 0, 5, 10, 15 ... 90, 95 ... 175, and 180 degrees as the recommended minimum. Preferably, 0, 2.5, 5.0, 7.5, 10 ... 90, 92.5 ... 177.5, and 180 degrees can be used. Angular increments of less than 2.5 degrees may be needed to adequately characterize uneven or sharply focused beam distributions. Such distributions may exist with lamps having a beam angle of less than 30 degrees. However, in no case shall the vertical angle steps exceed 10 degrees.

Luminous intensity values at each given vertical angle are averaged for the series of planes to obtain the mean readings at each vertical angle. These readings are multiplied by the appropriate zonal constants (zonal factors) to obtain the relative flux in each zone, and the values are summed to determine the total relative flux. From this total, the lamp factor may be calculated:

$$\text{Lamp Factor} = (\text{Lamp Rated Flux})/(\text{Total Relative Flux})$$

Luminaire luminous intensity values are determined by applying the lamp factor to subsequent readings taken when the complete luminaire is under test. These are values that would have resulted if the test lamp had

been delivering the published rated luminous output. Lamp electrical stabilization shall be maintained as defined in **Section 4.5**. Power, voltage, and current shall be checked with a calibrated instrument having an accuracy of ± 0.255 .

Candela readings taken on a goniophotometer can be converted into lumens using zonal constants. **Annex A** presents two basic formulas for this conversion, and numerous pre-calculated zonal constants are tabulated to aid the conversion process.

6.1.2 Absolute Method Photometry. With absolute method photometry, standard lamps (sources) of known luminous intensity in a specified direction are set up in the photometer. Then the photometer's instrumentation readout is adjusted with a calibration factor to display the known luminous intensity value of the standard lamp exactly. At least three standard lamps shall be compared in the test photometer, with the final calibration factor being an average of the calibration factors determined for each individual lamp. As previously mentioned, all electrical parameters shall be checked and held constant. The bare test lamp may be checked against a luminous flux standard to determine its lumen output.^{6,7}

NMI-based* absolutely calibrated detectors (rather than sources) may also be used.¹⁴ This newer method eliminates the need to maintain standard lamps. Alternative photometers may be calibrated by placing their photodetectors on an optical bench on the same illuminated plane as for a NMI-based photodetector.

This calibrated-detector method is claimed to have twice the accuracy of the older calibrated-source method. The NMI-based photodetector shall be calibrated for an illuminance range that includes the maximum illuminance level occurring in the beam of the luminaire under test.¹⁵

The lamp factor (F) thus becomes:

$$F = D^2/(RF)$$

where:

D = the test distance measured using the same system of units the detector uses (SI or USCS)

RF = the average spectral reflectance of the mirror in the visible region (if a mirror is used)

7.0 Photometric Measurement

7.1 Angles for Measurement

The vertical increments around the source or luminaire under test shall be small enough that the data taken describe the light distribution adequately. Vertical steps of 5 degrees are advised, but smaller increments (2.5° or smaller) shall be used if necessary. Vertical steps of 1° are advised whenever the beam angle is less than 30°. In no case shall vertical steps exceed 10°.

7.2 Planes for Measurement

The number of vertical planes explored shall be determined by the degree of asymmetry or irregularity existing within the light distribution being measured.

Where a lamp or luminaire is asymmetric, a series of vertical half-planes shall be used to describe the distribution accurately. At a minimum, readings should be taken in 16 vertical half-planes spaced 22.5° apart (see **Figure 7-1** and **Figure 7-2**). Highly asymmetric distributions may require a larger number of half-planes. In all cases, as many half-planes as are necessary to describe the photometric distribution adequately shall be taken.

For lamps or luminaires that have two planes of symmetry (are quadrilaterally symmetric), the five horizontal planes to the lamp axis are usually adequate (0° [parallel], 22.5°, 45°, 67.5°, and 90° [normal]) (see **Figure 7-1** and **Figure 7-2**). For bilaterally symmetric distributions (with one plane of symmetry), it will be necessary to report measurement planes from 0 through 180 degrees, with 22.5-degree steps usually adequate. The luminaire position should be clearly indicated with respect to the beam position for luminaires such as wall washers. For electronic published data, it is recommended that the 0-degree position be on the beam side.

Luminaire "raw data" measurements should be made in all four quadrants, at whatever lateral step size is

* NMI = national meteorological institute

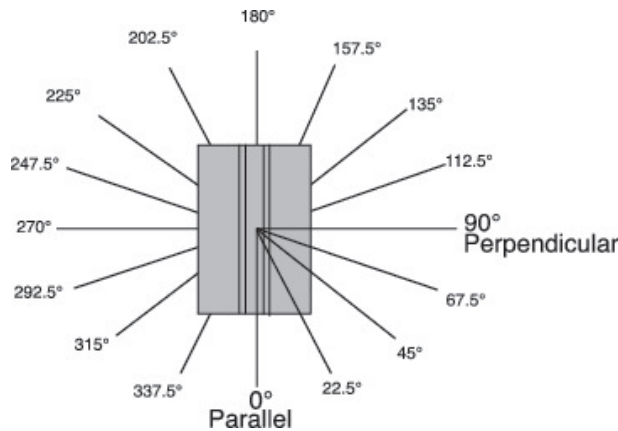


Figure 7-1. Horizontal positions for photometric half-planes recorded for luminaires (plan view).

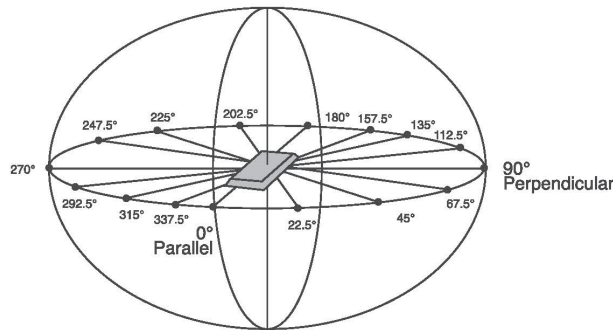


Figure 7-2. Horizontal positions for photometric half-planes recorded for luminaires (isometric view).

deemed adequate. The raw-data measurements should be averaged to present the desired reporting angles according to the distribution symmetry. For example, for quadrilateral symmetry, the following planes would be averaged together:

0°, 180°
22.5°, 157.5°, 202.5°, 337.5°
45°, 135°, 222.5°, 315°
67.5°, 112.5°, 247.5°, 292.5°
90°, 270°

For bilateral symmetry (plane of symmetry 0° to 180°), the following planes would be averaged together:

22.5°, 337.5°
45°, 315°
67.5°, 292.5°
90°, 270°

112.5°, 247.5°
135°, 225°
157.5°, 202.5°

For luminaires having a distribution that is axially symmetric about the vertical axis, readings shall be taken in at least ten, and preferably more, vertical half-planes (i.e., each in one sweep from nadir to zenith) and then averaged. Normally, 16 half-planes spaced 22.5° apart will be satisfactory.

8.0 Indirect Luminaires

Many luminaires are manufactured with their primary distribution of light between 90 degrees vertical and zenith. For accuracy, such indirect luminaires may require special photometric equipment that allows the luminaire to perform in its normal operating position during testing.

9.0 Reporting Test Data

Test reports should describe all significant data concerning the luminaire under test. The recorded data should at least provide the following information:

a. Luminaire Description

1. Manufacturer's name; type and catalog number of luminaire
2. Type of mounting
3. Reflectance (of painted surfaces and louvers)
4. Description of enclosing media, louvers (including surface finish, texture, and specularly)
5. Sketch showing luminaire shape, dimensions, and light center position(s)
6. Ballast data: single- or multiple-lamp, power factor correction, voltage, frequency, lamp current (where applicable), and catalog number of ballast(s)
7. Lamp watts, input watts, and input voltage (where applicable)

8. Luminaire Efficacy Rating (optional)¹⁶
- b. Test lamp description
 1. Number, type, bulb designations
 2. Rated watts
 3. Rated lumen output
- c. General lighting characteristics
 1. Luminous intensity distribution curves in designated planes
 2. Tabulated luminous intensity values for each plane in 5-degree steps (2.5-degree steps when such are deemed useful in interpreting and using the data)
 3. Tabulations of lumens and percentages of bare lamp lumens for zones 0 to 30 degrees, 0 to 40 degrees, 0 to 60 degrees, 0 to 90 degrees, 90 to 180 degrees, and 0 to 180 degrees; (for totally indirect luminaires, tabulation should be in zones 90 to 120, 90 to 130, 90 to 150, and 90 to 180 degrees)
 4. Curves or tabulations of average luminance data
 5. Test distance
- d. Date and testing agency
- e. Description: A complete description of any item(s) or procedure(s) peculiar to the luminaire being tested and/or its report.
- f. Statement: Include a statement: "This photometric test was performed using a specific ballast-lamp combination. Extrapolation of these data for other ballast-lamp combinations may produce erroneous results."
- g. Statement: Include a statement: "This photometric test was conducted in controlled laboratory conditions; e.g., ambient temperature was held at 25 °C ± 5 °C (77 °F ± 9 °F). Luminaire field performance may differ, particularly with regard to light output. Such changes result from different ambient temperatures and the way the luminaire is mounted."

Note: Statements (f) and (g) are provided to warn the user about particular factors involved in producing a photometric test report, which may change under actual applied conditions. Equivalent statements may

be used individually (or in combination) to best suit the needs of the agency preparing the test report.

Annex B provides a sample indoor photometric report.

10.0 Use of Computers and the IES Standard Format

A variety of computer programs are available that will perform calculations and create photometric reports when data are obtained as outlined in this Approved Method. Examples of computer calculation results include isoilluminance diagrams and coefficients of utilization. Computer software is recommended as a more efficient and effective method for manipulating photometric data. The basis for these programs is luminaire luminous intensity distribution data presented in the IES standard format.¹⁷ (See **Annex C** for an example of an IES formatted file.)

Annex A – Converting Candelas into Lumens Using Zonal Constants

This annex provides a method for converting candelas into lumens when candela readings are taken on a goniophotometer. For additional aid and convenience, a list of pre-calculated Zonal Constants is given in **Table A-1**. More detailed information about the following formulas and the use of **Table A-1** is available.^{18,19}

The Zonal Constant (K_{ω}) is determined from:

$$K_{\omega} = 2\pi(\cos\theta_1 - \cos\theta_2)$$

where:

K_{ω} = Zonal Constant, equal to solid angle of zone n

θ_1 = lower angular limit of zone n

θ_2 = upper angular limit of zone n

The zonal flux (ϕ_n) is determined from:

$$\phi_n = K_{\omega}I$$

where:

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Discharge or Incandescent Filament Lamps**

ϕ_n = flux in zone n (in lumens)

zone n (360-degree average about the polar axis)

I = luminous intensity (in candelas) at the center of

K_ω = Zonal Constant, equal to solid angle of zone n

**Table A-1. Zonal Constants Tabulated for Use in the Zonal Method of Computing Luminous Flux from
Candlepower Data**

1-Degree Zones		2-Degree Zones		5-Degree Zones		10-Degree Zones	
Zonal Limits (degrees)	Zonal Constant	Zonal Limits (degrees)	Zonal Constant	Zonal Limits (degrees)	Zonal Constant	Zonal Limits (degrees)	Zonal Constant
0 – 1	0.0009	0 – 2	0.0038	0 – 5	0.0239	0 – 10	0.095
1 – 2	0.0029	2 – 4	0.0115	5 – 10	0.0715	10 – 20	0.283
2 – 3	0.0048	4 – 6	0.0191	10 – 15	0.1186	20 – 30	0.463
3 – 4	0.0067	6 – 8	0.0267	15 – 20	0.1649	30 – 40	0.628
4 – 5	0.0086	8 – 10	0.0343	20 – 25	0.2097	40 – 50	0.774
5 – 6	0.0105	10 – 12	0.0418	25 – 30	0.2531	50 – 60	0.897
6 – 7	0.0124	12 – 14	0.0493	30 – 35	0.2946	60 – 70	0.993
7 – 8	0.0143	14 – 16	0.0568	35 – 40	0.3337	70 – 80	1.058
8 – 9	0.0162	16 – 18	0.0641	40 – 45	0.3703	80 – 90	1.091
9 – 10	0.0181	18 – 20	0.0714	45 – 50	0.4041		
					0.4349		
					0.4623		
					0.4862		
					0.5064		
					0.5228		
					0.5351		

Table note: Values were calculated using the Zonal Constant (K_ω) formula and shown for convenience.

Annex B – Sample Indoor Photometric Report

This sample report is based on ANSI/IES LM-46-20 and other pertinent IES procedures.

TEST NUMBER: 1234DATE: 00/00/00

PREPARED FOR: XYZ LIGHTING COMPANY

CATALOG NUMBER: XYZ 123

LUMINAIRE: FORMED ALUMINUM HOUSING, SPUN FLUTED SPECULAR ALUMINUM REFLECTOR, NO ENCLOSURE.

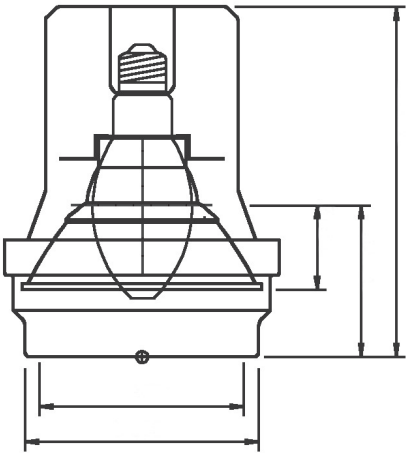
LAMP: ONE CLEAR 70-WATT ED17 3K METAL HALIDE RATED AT 5900 LUMENS.

BALLAST: ONE ELECTRONIC BALLAST

MOUNTING: TRACK

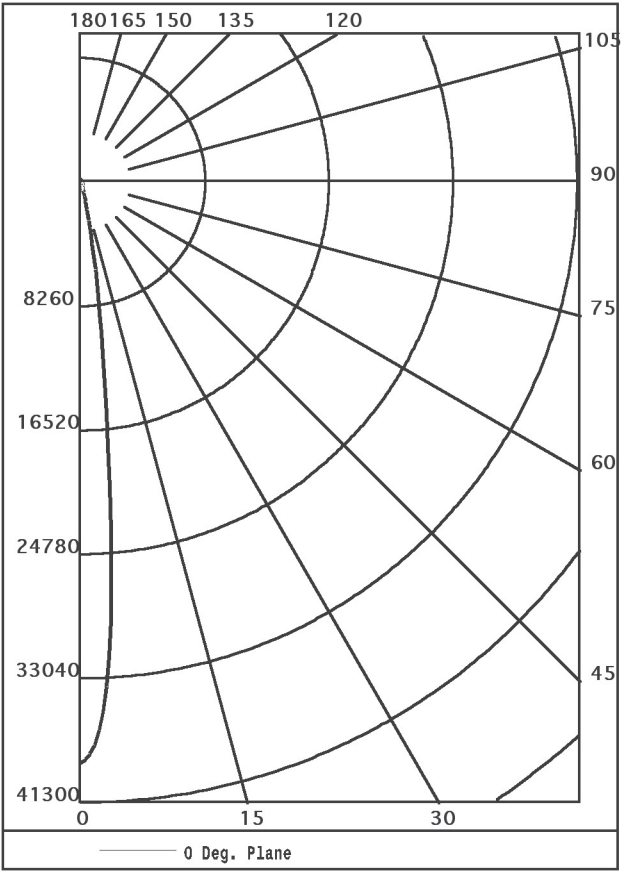
TOTAL INPUT WATTS: 78.1 AT 120.0 VOLTS

CANDELA DISTRIBUTION		
	0.0	FLUX
0	38807	
5	23465	1601
15	1322	466
25	182	96
35	38	24
45	0	0
55	0	0
65	0	0



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75	0	0
85	0	0
90	0	
ZONAL LUMEN SUMMARY		
ZONE	LUMENS	%LAMP
0- 30	2163	36.65
0- 40	2187	37.07
0- 60	2187	37.07
0- 90	2187	37.07
90-180	0	0.00
0-180	2187	37.07
TOTAL LUMINAIRE EFFICIENCY: 37.1%		
CIE TYPE: DIRECT		
SPACING CRITERION: 0.23		
LUMINOUS DIAMETER: 4.375		
LUMINANCE IN CANDELAS PER SQUARE METER:		
ANGLE	AVERAGE	
IN DEG	0-DEG	
0	4001664	
45	0	
55	0	
65	0	
75	0	
85	0	



COEFFICIENTS OF UTILIZATION - ZONAL CAVITY METHOD
EFFECTIVE FLOOR CAVITY REFLECTANCE OF 0.20

RC	90				80				70			50			10			0
RW	70	50	30	10	70	50	30	10	50	30	10	50	30	10	50	30	10	0
0	44	44	44	44	43	43	43	43	41	41	41	39	39	39	38	38	38	37
1	43	42	42	41	42	41	41	41	40	40	39	39	38	38	38	37	37	37
2	42	41	40	39	41	40	39	39	39	39	38	38	38	37	37	37	37	36
3	41	40	39	38	40	39	38	38	38	38	37	38	37	37	37	36	36	36
4	40	39	38	37	40	38	37	37	38	37	36	37	36	36	37	36	36	35
5	39	38	37	36	39	38	37	36	37	36	36	37	36	36	36	36	35	35

6	39	37	36	36	38	37	36	35	37	36	35	36	36	35	36	35	35	35
7	38	37	36	35	38	36	36	35	36	35	35	36	35	35	36	35	34	34
8	38	36	35	35	38	36	35	34	36	35	34	35	35	34	35	35	34	34
9	37	36	35	34	37	36	35	34	35	35	34	35	34	34	35	34	34	34
10	37	35	34	34	37	35	34	34	35	34	34	35	34	34	35	34	34	33

CANDELA DISTRIBUTION

ZONAL LUMEN SUMMARY

(1-degree steps)					
	0.0			0-5	755
0	38807	19	620	5-10	846
1	38174	20	595	10-15	339
2	36595	21	351	15-20	127
3	33731	22	257	20-25	64
4	29325	23	238	25-30	32
5	23465	24	219	30-35	24
6	17881	25	182	35-40	0
7	13550	26	138	40-45	0
8	10228	27	150	45-50	0
9	7828	28	132	50-55	0
10	5979	29	138	55-60	0
11	4293	30	119	60-65	0
12	3215	31	119	65-70	0
13	2206	32	88	70-75	0
14	1711	33	69	75-80	0
15	1322	34	69	80-85	0
16	1065	35	38	85-90	0
17	702	36	25		
18	727	37-90	0		

THIS TEST WAS CONDUCTED USING RELATIVE PHOTOMETRY TECHNIQUES ACCORDING TO STANDARD IESNA PROCEDURES. THE USER MUST THEREFORE USE CAUTION IN THE FOLLOWING SITUATIONS: 1) ACCORDING TO IESNA PROCEDURES, THE BALLAST(S) AND LAMP(S) ARE PRESUMED TO PRODUCE 100% OF RATED OUTPUT. AN APPROPRIATE BALLAST FACTOR MUST BE APPLIED TO THE LUMEN OUTPUT RATINGS AND LUMINOUS INTENSITY VALUES GIVEN. 2) THIS TEST WAS CONDUCTED IN A CONTROLLED LABORATORY ENVIRONMENT, FIELD PERFORMANCE MAY DIFFER.

0. 0. 0. 0. 0. 0. 0. 0. 0.

Annex D – Sample Luminaire Simulator

The luminaire simulator diagram¹ shown in **Figure D-1** is for illustrative purposes only. Specific individual luminaire simulator configurations will vary.

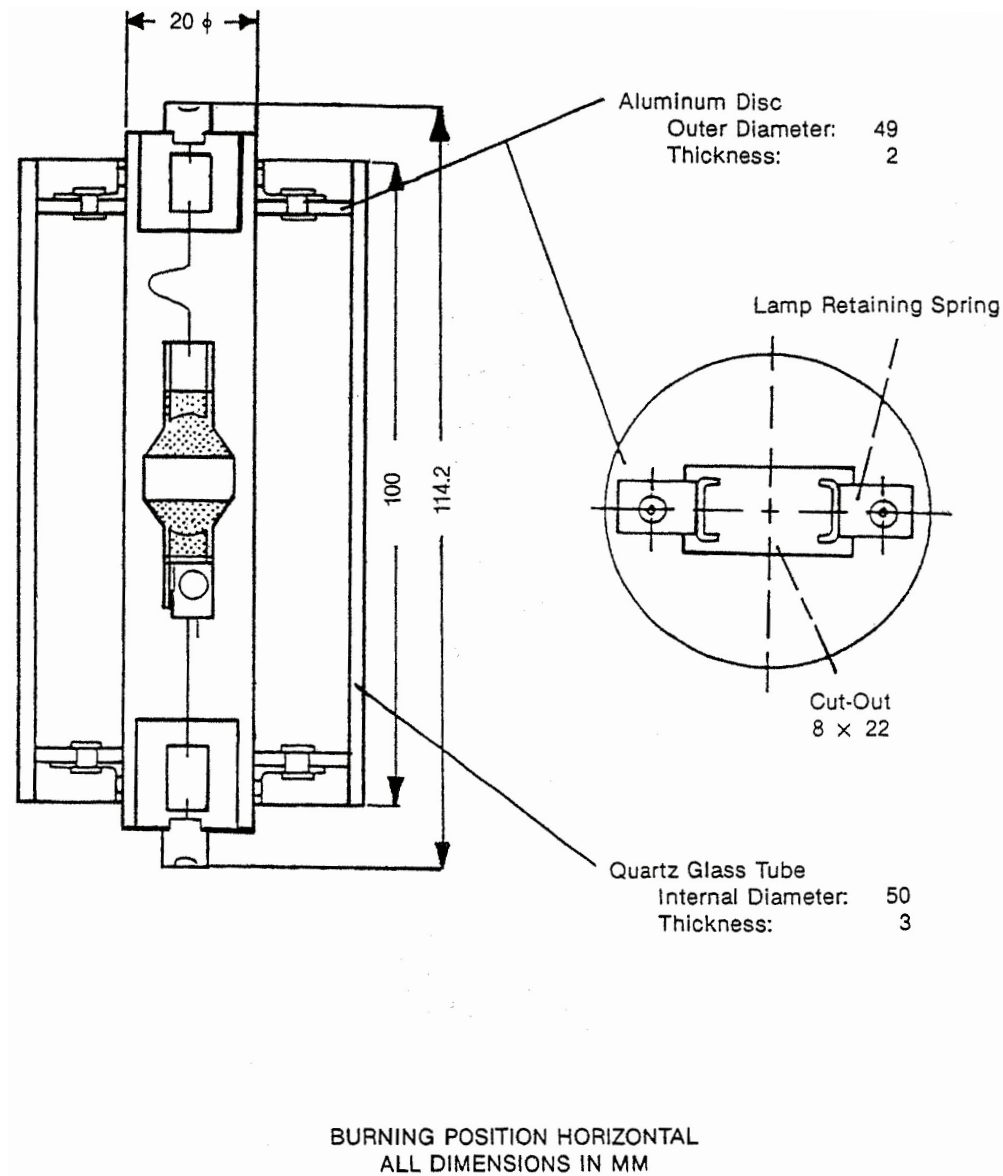


Figure D-1. Example of a luminaire simulator diagram. (Source: *HQI-TS 70 – 1000 Watt Fixture Design Guide*; OSRAM SYLVANIA , 1989.)

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ISBN# 978-0-87995-211-2

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